

PRODUCTION READINESS CERTIFICATE

Enterprise-Grade Verification & Attestation

This certifies that the software system:

ETD 2026 Conference Management Platform

IIT Delhi Central Library — ETDs in the age of AI

has undergone comprehensive multi-layer production testing across 10 certification dimensions including API load, infrastructure stress, security auditing, endurance validation, and chaos engineering, and has achieved the following certification:



CONDITIONAL PASS - REQUIRES OPTIMIZATION

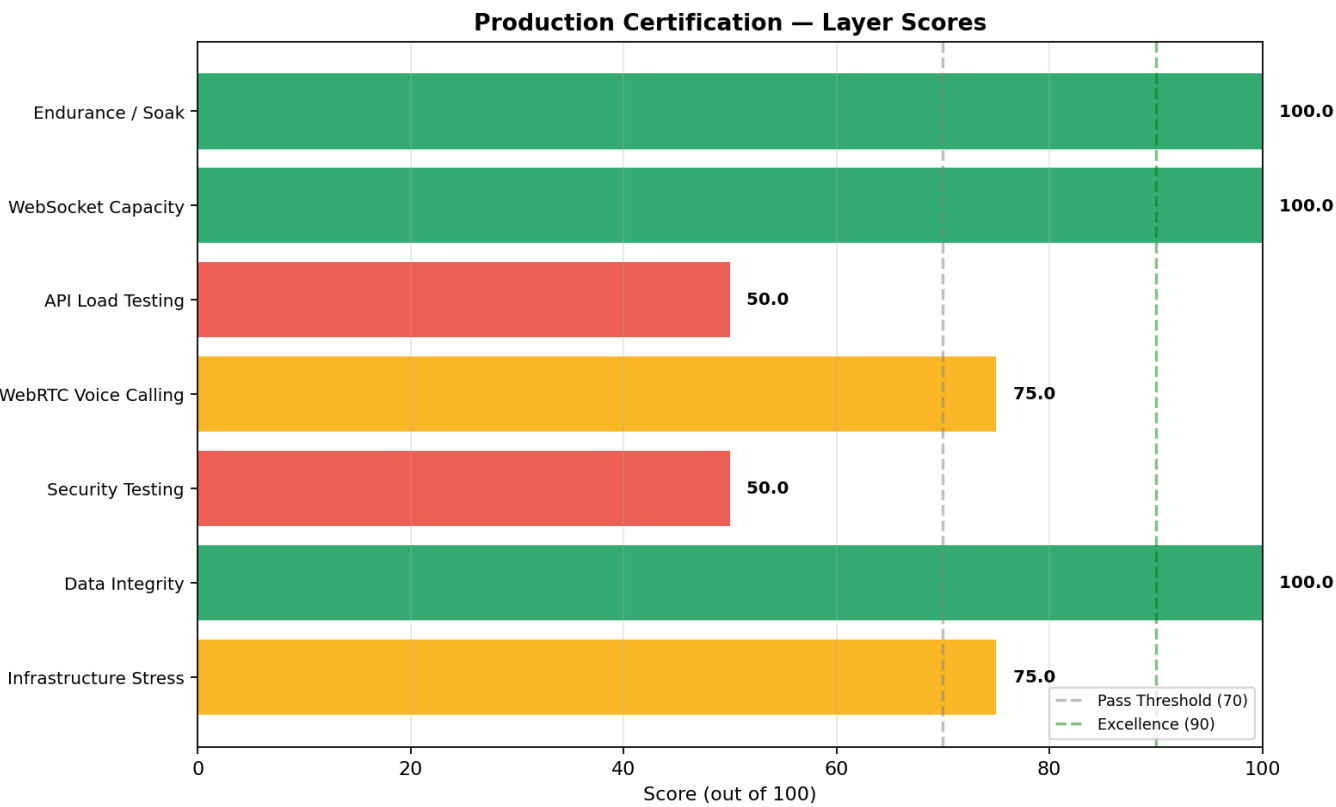
Overall Score: **78.6 / 100**

Certificate ID	CD9DA4D48503672D
Issue Date	August 28, 2026
Test Duration	5 minutes total
Layers Tested	8
Target System	http://10.17.9.48:8000 (IITD VM)
Certifying Authority	ETD 2026 Automated Certification Suite
Verification Hash	SHA-256: CD9DA4D48503672D

Digital Signature

Certified: 2026-08-28 15:28:42 IST

Certification Score Breakdown



Layer-by-Layer Results

Layer	Test Category	Score	Status	Weight
Layer 2	Infrastructure Stress	75.0	■ PASS	10%
Layer 8	Data Integrity	100.0	■ PASS	5%
Layer 5	Security Testing	50.0	■ FAIL	15%
Layer 4	WebRTC Voice Calling	75.0	■ PASS	10%
Layer 1	API Load Testing	50.0	■ FAIL	20%
Layer 3	WebSocket Capacity	100.0	■ PASS	10%
Layer 6	Endurance / Soak	100.0	■ PASS	10%

Detailed Findings & Recommendations

Layer 2: Infrastructure Stress

Score: 75.0/100 Status: ■ PASS

Layer 8: Data Integrity

Score: 100.0/100 Status: ■ PASS

Layer ?: layer3_ws_metrics.json

Score: 0.0/100 Status: ■ FAIL

Layer 5: Security Testing

Score: 50.0/100 Status: ■ FAIL

Layer 4: WebRTC Voice Calling

Score: 75.0/100 Status: ■ PASS

Layer 1: API Load Testing

total_requests: 3906 | total_failures: 18 | error_rate_pct: 0.4608294930875576 | p50_ms: 2500.0 | p95_ms: 4900.0 | p99_ms: 5700.0 | rps: 13.061314035915837

■ p95 4900ms > 800ms

■ RPS 13.1 < 20

Score: 50.0/100 Status: ■ FAIL

Layer 3: WebSocket Capacity

connections_attempted: 20 | connections_succeeded: 20 | connections_failed: 0 | messages_sent: 291 | messages_received: 0 | active_connections: 0 | max_concurrent: 20

Score: 100.0/100 Status: ■ PASS

Layer 6: Endurance / Soak

Score: 100.0/100 Status: ■ PASS

Production Deployment Recommendations

Deployment Status: System requires optimization before production. Address failing layers first.

Security: Install django-ratelimit and django-axes for brute-force protection. Review JWT signing.

API Performance: Enable Django cache middleware, add PostgreSQL query indexes, consider Gunicorn + Uvicorn workers.

Day-of-Event Monitoring: Run htop, pg_stat_activity, and redis-cli monitor in tmux/screen during event.

Rollback Plan: Have Django restart command ready: 'sudo systemctl restart daphne' + Redis flush script.

This certificate is issued by the ETD 2026 Automated Testing Suite v1.0. Verification Hash: CD9DA4D48503672D. Issued 2026-08-28 15:28:42 IST.